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11. IMPACT OF CULTURAL HERITAGE

11.1 Introduction

11.1.1 This section presents the Cultural Heritage Impact Assessment (CHIA) findings of the Project. The CHIA, which covers Built Heritage Impact Assessment (BHIA) and Archaeological Impact Assessment (AIA), has been conducted in accordance with the requirements given in Clause 3.5.11 and Appendix J of the EIA Study Brief (No. ESB-375/2025).

11.1.2 The assessment area for CHIA is 300m from the Project Boundary (**60722948/A10/FIGURE 11.1** refers). Built heritage and archaeological heritage within the 300m assessment area have been identified and the potential direct and indirect impacts resulting from the construction and operation of the Project on the identified built heritage and archaeological heritage have been assessed. Appropriate mitigation measures have been proposed to alleviate the adverse impacts as necessary.

11.2 Environmental Legislations, Standards and Guidelines

11.2.1 Legislation, standards and guidelines relevant to the CHIA include the following:

- Environmental Impact Assessment Ordinance (EIAO) (Cap. 499)[1];
- Annexes 10[2] and 19[3] of Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM); and
- Antiquities and Monuments Ordinance (A&MO) (Cap. 53)[4].

Environmental Impact Assessment Ordinance (EIAO) (Cap. 499) and EIAO-TM Annexes 10 and 19

11.2.2 The EIAO was implemented on 1st April 1998. It aims to avoid, minimise and control the adverse impacts on the environment of designated projects, through the EIA process and the Environmental Permit (EP) system.

11.2.3 According to Schedule 1 of the EIAO, Site of Cultural Heritage refers to "an antiquity or monument, whether being a place, building, site or structure or a relic, as defined in the Antiquities and Monuments Ordinance (Cap. 53) and any place, building, site, or structure or a relic identified by the Antiquities and Monuments Office to be of archaeological, historical or palaeontological significance."[5]

11.2.4 Annexes 10 and 19 of EIAO-TM provide the criteria and guidelines for evaluating the impacts to Sites of Cultural Heritage. It is stated in Annex 10 that all adverse impacts to Sites of Cultural Heritage shall be kept to an absolute minimum and that the general presumption of impact assessment shall be in favour of the protection and conservation of all Sites of Cultural Heritage. Annex 19 provides the commonly

[1] Hong Kong e-Legislation. *Cap. 499 Environmental Impact Assessment Ordinance*. Retrieved from https://www.elegislation.gov.hk/hk/cap499.
[2] Environmental Protection Department. *Annex 10: Criteria For Evaluating Landscape And Visual Impact, And Impact On Sites Of Cultural Heritage*. Retrieved from https://www.epd.gov.hk/eia/en/legis/memorandum/annex10.html
[3] Environmental Protection Department. *Annex 19: Guidelines For Assessment Of Impact On Sites Of Cultural Heritage And Other Impacts*. Retrieved from https://www.epd.gov.hk/eia/en/legis/memorandum/annex19.html
[4] Hong Kong e-Legislation. *Cap. 53 Antiquities and Monuments Ordinance*. Retrieved from https://www.elegislation.gov.hk/hk/cap53.
[5] Hong Kong e-Legislation. Cap. 499 Environmental Impact Assessment Ordinance. Retrieved from https://www.elegislation.gov.hk/hk/cap499?xpid=ID_1438403274391_002.

adopted approaches and methodologies for assessment of impact on Sites of Cultural Heritage.

Antiquities and Monuments Ordinance (A&MO) (Cap. 53)

11.2.5 The *Ordinance* provides the statutory framework for preservation of objects of historical, archaeological and paleontological interest and for matters ancillary thereto or connected therewith. The *Ordinance* contains the statutory procedures for the Declaration of Monuments. Under the *Ordinance*, a "monument" means a place, building, site or structure which is declared to be a monument, historical building or archaeological or paleontological site or structure under Section 3 of the *Ordinance*. Excavations carried out on building works, demolition and interference of a proposed monument or monument are prohibited except under permit under Section 6 of the *Ordinance*.

11.3 Assessment Methodology

11.3.1 Following the requirement Clause 3.5.11 and Appendix J of the EIA Study Brief (No. ESB-375/2025), the CHIA consists of a BHIA and an AIA. The assessment methodology for BHIA and AIA are described below.

Built Heritage Impact Assessment (BHIA)

11.3.2 A desktop review was conducted to identify any built heritage within 300m assessment area based on examination on the following resources:

- List of declared monuments and proposed monuments[6];
- List of the graded historic buildings[7]
- List of new items for grading assessment[8];
- List of Government historic sites Identified by Antiquities and Monuments Office[9];

11.3.3 Site visits were carried out in the assessment area on 29 April 2024 to evaluate the current condition of built heritage identified during the desktop review, as well as any items that might not be revealed by the desktop review.

11.3.4 The potential direct and indirect impacts on the built heritage during the construction and operation phases of the Project have been assessed in the CHIA by following the procedures and requirements of Annexes 10 and 19 of the EIAO-TM.

11.3.5 Mitigation measures are proposed in the CHIA for any affected built heritage to minimise any adverse impacts when necessary.

[6] Antiquities and Monuments Office. Declared Monuments in Hong Kong (as at 10 October 2024). Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/DM_Mon_List.pdf
[7] Antiquities Advisory Board. List of the 1,444 Historic Buildings with Assessment Results (as of 10 October 2025). Retrieved from https://www.aab.gov.hk/filemanager/aab/en/content_29/AAB-SM-chi.pdf
[8] Antiquities Advisory Board. List of New Items for Grading Assessment with Assessment Results (as of 11 September 2025). Retrieved from https://www.aab.gov.hk/filemanager/aab/en/content_29/list_new_items_assessed.pdf
[9] Antiquities and Monuments Office. *Government Historic Sites Identified by AMO (as at May 2022)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/build_hia_government_historic_sites.pdf.

Archaeological Impact Assessment (AIA)

11.3.6 A desktop review was conducted to identify any potential existence of archaeological heritage based on examination on the following resources, when available: -

- List of Sites of Archaeological Interest[10] identified by the AMO;
- Previous related EIA studies and archaeological reports;
- Related publications and monographs on relevant archaeological, historical and geographical issues;
- Unpublished archival papers and records, and collection and libraries of tertiary institutions; and
- Geological and historical maps, aerial photos and relevant visual archives.

11.3.7 The potential impacts that may affect the possible archaeological heritage during the construction and operational phases of the Project have been assessed in the CHIA by following the procedures and requirements of Annexes 10 and 19 of the EIAO-TM.

11.3.8 In case adverse impacts on archaeological heritage cannot be avoided, appropriate mitigation measures would be designed and recommended in the CHIA to minimise the impacts.

11.4 Project Background

Geographical and Geological Background

Physical Geography

11.4.1 The Project Boundary is located in Lok Ma Chau, New Territories, which is to the south of Shenzhen River. It starts at the foot of *Tai Shek Mo* (大石磨; also known as Crest Hill), running along the north of the hill of Lok Ma Chau, and ends at the Loop.

11.4.2 Currently, the terrestrial elevation within the Project Boundary is generally ranged between +3.5mPD to +22.0mPD. Its gradient ranges from <1° to 10° in plains, and gradually increasing to <30°.

Past Landscape

11.4.3 Sea level rose approximately by 1m around 6,500 BP [11]. After a rise in sea level, there was a drop in sea level approximately by 3m around 6,000 BP[12] [13]. The plain ("QHH" and "Qam") at the south of Shenzhen River was formed by sedimentation less than 2,000 years ago[14].

11.4.4 According to the geology information, there were river valleys in the vicinity of the Loop. The river valley was primarily shaped by *Carboniferous Metamorphosed*

[10] Antiquities and Monuments Office. *List of Sites of Archaeological Interest in Hong Kong (as at Nov 2012)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/list_archaeolog_site_eng.pdf.
[11] 李平日、方國祥、黃光慶。（1996年6月）。＜珠江三角洲全新世環境演變＞。《第四紀研究》。2，135。
[12] Fyfe, J. A et al. (2000). *The Quaternary Geology of Hong Kong*. Hong Kong: Civil Engineering Department.
[13] 李平日、方國祥、黃光慶。（1996年6月）。＜珠江三角洲全新世環境演變＞。《第四紀研究》。2，135。
[14] Civil Engineering and Development Department & Planning Department. (2013). Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report. Civil Engineering and Development Department & Planning Department.

Sedimentary rocks ("Cts" and "Cmp"), which formed the hilly landscape to the east and south of the Project Boundary. Debris flow deposits ("Qpd") were formed along the small hills. The presence of a river running through the region led to the formation of the second river terrace ("Qpa"), while a drop in sea level around 6,000 BP resulted in the formation of the first river ("Qa")[15]. Details can be referred to **60722948/A10/FIGURE 11.2**.

Human Geography

11.4.5 Three villages are located in the vicinity of the Project Boundary, namely *Ma Tso Lung Shun Yee San Tsuen* (馬草壟信義新村), *Ma Tso Lung San Tsuen* (馬草壟新村) and *Tse Koo Hang* (鷓鴣坑). *Ma Tso Lung Shun Yee San Tsuen* (also known as *Ma Tso Lung Lutheran New Village*) is located to the west foot of *Tai Shek Mo*. Meanwhile, *Ma Tso Lung San Tsuen* is to the south of *Ma Tso Lung Shun Yee San Tsuen*. On the other hand, *Tse Koo Hang* is located in a river valley to the south of the Project Boundary. According to the aerial photo in 1924, part of *Ma Tso Lung Shun Yee San Tsuen* and *Tse Koo Hang* were agricultural lands. In 1962[16], *Ma Tso Lung Shun Yee San Tsuen* was established. *Ma Tso Lung San Tsuen* (馬草壟新村) was constructed between 1973[17] and 1974[18]. *Tse Koo Hang* was established between 1950s and 1970s according to the aerial photo[19] and map[20].

Historical Background

Qin to Yuan Dynasties (221BC-AD1368)

11.4.6 Clues of human occupation within the south China can be found in historic textual records such as *Shiji* (史記) and *Hanshu* (漢書), written in the first century BC and first century AD respectively. These records describe that *Yue* ethnic groups (also known as *Hundreds of Yue* (百越)) were scattered in southern China. The *Yue* ethnic groups were comprised of different tribes bearing various surnames and can be differentiated from the *Han* ethnic group who lived in central China in terms of physical characteristics, language, and folklore.

11.4.7 The *Yue* people were gradually assimilated into the *Han* culture when southern China became an administration territory of the central government since *Qin* dynasty (221-206BC). During the *Qin* period, the *Guangdong* region was subordinated to *Panyu* (番禺) County. In 208 BC, *Southern Yue State* (南越國) was established around the *Guangdong* region by military officials, who were sent from the *Qin* Court to conquer the *Yue* in the south. Following the collapse of *Qin*'s political power in the north, *Han* dynasty (206BC-AD220)[21] began. *Southern Yue State* was soon becoming a vassal state of *Han* before integrated into the *Han* Empire.

11.4.8 Between *Han* and *Eastern Jin* dynasties (AD317-420), Hong Kong was subordinated to *Bolou* (博羅) County[22]. From AD331 to AD756, Hong Kong was subordinated to

[15] Fyfe, J. A et al. (2000). *The Quaternary Geology of Hong Kong*. Hong Kong: Civil Engineering Department.
[16] 華僑日報（1962年11月29日）。＜馬草壠信義新村今午舉行開幕禮＞。《華僑日報》。
[17] Survey and Mapping Office, Lands Department. (1973). *Topographic Map in 1973 [map]. 1:1,200. 58-NE-D.* Survey and Mapping Office, Lands Department.
[18] Survey and Mapping Office, Lands Department. (1974). *Topographic Map in 1974 [map]. 1:1,200. 58-NE-D.* Survey and Mapping Office, Lands Department.
[19] Survey and Mapping Office, Lands Department. (1973). *Aerial Photos [aerial photos]. 1:25,000. (12,500 ft.) 07857R.* Survey and Mapping Office, Lands Department.
[20] War Office. (1952). *Topographic Map in 1952 [map].* War Office.
[21] 司馬遷 (c.a. 91BC)。史記 卷一百一十三 南越列傳 第五十三。北京：中華書局 (1959) 。
[22] Although the boundary between Boluo (博羅) County and Panyu (番禺) County during Han to East Jin period is unclear, it is generally suggested that Hong Kong region belonged to Boluo County at that time, according to Xinan Gazetteer (1819), Social

Bao'an (寶安) County. After AD757, Hong Kong was subordinated to *Dongguan* (東莞) County and followed by *Song* dynasty (AD960-1279) and *Yuan* dynasty (AD1271-1368) [23].

Ming to Qing Dynasties (AD1368-1912)

11.4.9 During the 15th century, the coastal areas of *Dongguan* County suffered from frequent marauding bandit and pirate attacks. *Xin'an* (新安) County was thus set up in AD1573 to defend such attacks. According to *Xin'an Gazetteer* (新安縣誌)[24], the modern region of Hong Kong fell within the *Xin'an* County.

11.4.10 In 1661, Coastal Evacuation Order was compelled by the *Qing* Court in order to stifle the anti-Manchu troops in Taiwan. People living in coastal area were forced to move 50 *li* (里) (approximately 25 km) inland, including the New Territories inhabitants. The Order was lifted in 1669. However, after the coastal evacuation, population dropped severely. During the *Shunzhi* reign (1643-1661), the population of *Xin'an* County was recorded as 6,851. The population dropped to 2,172 in 1664 during the enforcement of the Order. After the Order was lifted, people were encouraged to move back to *Xin'an* County during late 17th to early 18th centuries. In 1671, the population increased to 3,972, 1,648 people were encouraged to move back during 1669-1671[25].

11.4.11 After the First Opium War (1839-1842) between the *Qing* government and the British Empire, the *Qing* government "…ceded … the Island of Hongkong, to be possessed in perpetuity by … Great Britain" signed in 1842 under the *Treaty of Peace, Friendship, and Commerce* Between *Her Majesty The Queen of Great Britain and Ireland and the Emperor of China* (also known as the *Treaty of Nanking* (南京條約)) [26]. The *Qing* government lost the Second Opium War (1856-1860), which led to the ceding Kowloon as a dependency of Hong Kong under the *Convention of Peace Between Her Majesty and The Emperor of China* (also known as the Convention of Peking (北京條約)) in 1860[27]. At the turn of the 20th century, *The Convention between the United Kingdom and China, Respecting an Extension of Hong Kong Territory* (also known as the Second Convention of Peking (第二北京條約)) signed between the British and the Qing government in 1898 allowed the British colony to "… enlarged under lease … [for] ninety-nine years." [28]

11.4.12 The Loop has been part of Shenzhen before the training of Shenzhen River. On 1st July 1997, it was announced that the new border between Hong Kong and Shenzhen would be the centreline of Shenzhen River after the river training works[29]. The Shenzhen River Regulation Project was completed in 2017[30]. In the early 20th century, the settlements primarily consisted of loosely clustered temporary structures. These

Change in Hong Kong Before and After the Early Qing Clearance (1986), and Brief History of Ancient Shenzhen (1997). However, Professor Jao Tsung-I (2005) discussed that the area belonged to Panyu based on the inscriptions on bricks of Lei Cheng Uk Han Tomb.

[23] 劉智鵬、劉蜀永 (編) (2020)《方志中的古代香港-《新安縣志》香港史料選》。香港，三聯書店（香港）有限公司。

[24] 劉智鵬、劉蜀永 (編) (2020)《方志中的古代香港-《新安縣志》香港史料選》。香港，三聯書店（香港）有限公司。

[25] 靳文謨修、鄧文蔚纂 (1688)。《新安縣志》。在廣東省地方志辦公室編，廣東歷代方志集成：廣州府部（二六）[康熙]新安縣志 [嘉慶]新安縣志。廣東：嶺南美術出版社。

[26] Mayers, William Fredrick. (1902, 4th edition). *Treaties Between the Empire of China and Foreign Powers*. Shanghai: North-China Hera Treaty of London (1871).

[27] *Ibid.*

[28] *Ibid.*

[29] 香港特別行政區政府及深圳市人民政府 (2017年1月3日)。《香港特別行政區政府 深圳市人民政府 關於港深推進落馬洲河套地區共同發展的合作備忘錄》。https://gia.info.gov.hk/general/201701/03/P2017010300605_250711_1_1483450812115.pdf.

[30] Drainage Services Department. *Shenzhen River Regulation Project*. Drainage Services Department.

settlers were predominantly migrants from Shenzhen (深圳) who chose to make their home in this region[31].

11.4.13 These settlers in Ma Tso Lung initially settled on river plains near the farmlands, which occasionally subjected them to flooding during rainy seasons. In the early 1960s, the formerly *Swedish National Committee of Lutheran World Federation* (瑞典世界信義宗) (now known as Hong Kong Christian Service) recognised the poor living conditions of the settlers. As a result, this organisation made donation and constructed houses in Ma Tso Lung to improve the living standards of the settlers. In 1962, with the assistance of the North District Office and the Sheung Shui District Rural Committee[32], a new village was named *Ma Tso Lung Shun Yee San Tsuen* was established in gratitude towards the *Swedish National Committee of Lutheran World Federation*.

11.4.14 According to topographic maps, a cluster of buildings currently known as *Ma Tso Lung San Tsuen* (馬草壟新村) was constructed between 1973[33] and 1974[34]. It is believed that these settlements were built by the settlers from *Ma Tso Lung Shun Yee San Tsuen* and some immigrants from *Dongguan[35]*. Meanwhile, *Tse Koo Hang* was established between 1950s and 1970s according to the aerial photo[36] and map[37].

11.4.15 In the late 20th century, an increase in the number of temporary structures scattered within the river valley can be observed[38]. These structures overlooked the surrounding farmlands, reflecting the ongoing agricultural activities in the region.

11.4.16 The geographical location of Lok Ma Chau is not only suitable for human settlements but also border security. In 1951, the Frontier Closed Area was established under the *Frontier Closed Area Order* to combat illegal migration and smuggling activities across the border[39]. This action was taken in response to suspicious that some border villages were smuggling equipment to mainland to assist Chinese armies during the Korean War.[40] As a result, the northern part of Ma Tso Lung, including Shun Yee San Tsuen and Ma Tso Lung San Tsuen, was included within the closed area.

11.4.17 Between 1949 and 1953, seven (7) observation posts were strategically constructed along the territory border on hilltops to strengthen border defence[41]. These observation posts later known as MacIntosh Forts, were named after Duncan MacIntosh, the Commissioner of Police, who authorized the construction.[42] In

[31] 華僑日報(1962年11月30日)。＜張人龍致詞居民獲得新生從此安居樂業＞。《華僑日報》。
[32] 華僑日報 (1962年11月29日)。＜馬草壠信義新村今舉行開幕禮＞。《華僑日報》。
[33] Survey and Mapping Office, Lands Department. (1973). *Topographic Map in 1973 [map]. 1:1,200. 58-NE-D.* Survey and Mapping Office, Lands Department.
[34] Survey and Mapping Office, Lands Department. (1974). *Topographic Map in 1974 [map]. 1:1,200. 58-NE-D.* Survey and Mapping Office, Lands Department.
[35] 阮志 (2016)。《越界:香港跨境村莊及文化遺產》。香港，三聯書店（香港）有限公司。
[36] Survey and Mapping Office, Lands Department. (1973). *Aerial Photos [aerial photos]. 1:25,000. (12,500 ft.) 07857R.* Survey and Mapping Office, Lands Department.
[37] War Office. (1952). *Topographic Map in 1952 [map].* War Office.
[38] Survey and Mapping Office, Lands Department. (2005). [aerial photo] 1:5,000, 2,500 ft. CW64054. Survey and Mapping Office, Lands Department.
Survey and Mapping Office, Lands Department. (2005). [aerial photo] 1:5,000, 2,500 ft. CW65974. Survey and Mapping Office, Lands Department.
[39] 阮志 (2014)。《入境問禁: 香港邊境禁區史》。香港，三聯書店（香港）有限公司。
[40] 阮志 (2012)。《中港邊界的百年變遷:從沙頭角麻坑村說起》。香港，三聯書店（香港）有限公司。
[41] Antiquities Advisory Board (2010). Historic Building Appraisal – MacIntosh Fort (Ma Tso Lung). Antiquities Advisory Board https://www.aab.gov.hk/filemanager/aab/common/historicbuilding/en/303_Appraisal_En.pdf
[42] *Ibid*.

particular, MacIntosh Fort (Ma Tso Lung) was located within the assessment area, serving as a crucial component of the border surveillance system.

Archaeological Background

11.4.18 There is no Site of Archaeological Interest (SAI) identified within 300m assessment area. Based on desktop review, several archaeological surveys were conducted by others within Project Boundary and 300m assessment area. Details of the previous archaeological surveys can be referred to **60722948/A10/FIGURE 11.3a**.

Drainage Improvement in the Northern New Territories - Package B Archaeological Survey Final Report[43]

11.4.19 In 2003, an archaeological survey was carried out as part of the *Drainage Improvement in the Northern New Territories - Package B* Project. The project involved the construction of drainage channels in Ma Tso Lung[44], which is partially located within the Project Boundary of this Project. A total of fifteen (15) test pits and one hundred and sixty-four (164) auger holes were excavated in Northern New Territories. Among the test pits and auger holes abovementioned, two (2) test pits and sixteen (16) auger holes were excavated in Ma Tso Lung area, of which two (2) test pits and nine (9) auger holes are located within the Project Boundary of and 300m assessment area of this Project.

11.4.20 Archaeological findings of *Tang* period kiln debris and pottery fragments dated to *Han* and early *Tang* dynasty were found in the field survey, which is outside the 300m assessment area of this Project (**60722948/A10/FIGURE 11.3a** refers). The findings were "found as surface finds along the margin of two adjacent fish ponds on the alluvial plain at Ma Tso Lung" [45]. That survey concluded that "[t]*he broad-scale disturbance of this area and conversion from paddy fields to fish ponds in the past 40 years has obscured much of the original landscape. Such disturbance may however throw up the traces of settlements and activity during prehistory and the early historic period*…"[46]

11.4.21 A brief follow-up investigation is recommended to evaluate the nature and the extent of the surface archaeological material found during this field work. With reference to the report[47], the evaluation should include:

- Field recording of the spatial distribution of surface finds at the margin of the fish pond
- Preparation of cleaned 1m x 2m cross sections at the edge of the fish ponds where pottery appears of highest concentration to establish (if any) horizontal layers or depositional history
- The results of steps 1 and 2 will provide a basis for a program of 4-5 test pits in the area of interpreted site focus

[43] Archaeo-Environments Ltd. (2004) Drainage Improvement in the Northern New Territories – Package B Agreement CE 5/2002(DS) Archaeological Survey Final Report. Archaeo- Environments Ltd.
[44] Drainage Services Department. (2003). *Drainage Services Department Annual Report 2002-2003 – Appendix A*. Drainage Services Department.
[45] Archaeo-Environments Ltd. (2004) Drainage Improvement in the Northern New Territories – Package B Agreement CE 5/2002(DS) Archaeological Survey Final Report, p.56. Archaeo- Environments Ltd.
[46] *Ibid.*
[47] Archaeo-Environments Ltd. (2004) *Drainage Improvement in the Northern New Territories – Package B Agreement CE 5/2002(DS) Archaeological Survey Final Report, p.57.* Archaeo- Environments Ltd.

- The test pits will be excavated within the bunds in this focus area and will provide detailed stratigraphy of the site

Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report (Application No.: EIA-212/2013)[48,49]

11.4.22 The report suggested that the fish ponds in the area only possess "low archaeological potential"[50]. "Fragments of Tang period kiln debris and pottery dated to Han and early Tang Dynasty (2nd to 7th Century AD) were found during field survey across an 80m arc at the edge of two adjacent fish ponds - located about 500m northeast of the LMC Loop area. The conversion from paddy fields to fish ponds in the past 40 years has resulted in disturbance which – in this case – appears to have thrown up archaeological remains from prehistory and the early historic period. Areas within the LMC Loop occupied by dumped sediment and marshland have low archaeological potential." [51]

11.4.23 Meanwhile, the low hill slopes at Ma Tso Lung area, including area in Shun Yee San Tsuen and Ma Tso Lung San Tsuen, were considered to possess "moderate archaeological potential" (**60722948/A10/FIGURE 11.3b** refers). The report suggested that "these areas have moderate archaeological potential as sites of prehistoric or early historic period occupation at the edge of the former shallow sea"[52].

11.4.24 Therefore, an archaeological survey was conducted under that project of the *Planning and Engineering Study on Development of Lok Ma Chau Loop* in 2012, which included these moderate archaeological potential areas. The survey including field scanning, auger hole test and test pit excavation was conducted in the area along Border Road, at Ma Tso Lung, Ma Tso Lung Road and Ho Sheung Heung Road. The surveyed areas were located in hill slopes, margin of fish ponds and alluvium. The then-proposed Eastern and Western Connection Roads were included in the survey area.

11.4.25 Sixty-six (66) auger holes and twenty-one (21) test pits were excavated in this survey, in which forty-two (42) auger holes and fifteen (15) test pits were located within the Project Boundary or 300m assessment area of this Project. In particular, the survey areas included 1) Border Road, 2) Ma Tso Lung and 3) Ma Tso Lung Road, which fall into the Project Boundary of this Project. In total, eighteen (18) test pits and fifty-two (52) auger holes were excavated in these survey areas. The survey findings did not identify any sign of archaeological potential (**Table 11.1** and **60722948/A10/FIGURE 11.3c** refer).

[48] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)
[49] Civil Engineering and Development Department & Planning Department. (2013). *Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report.* Civil Engineering and Development Department & Planning Department.
[50] Civil Engineering and Development Department & Planning Department. (2013). *Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report,* p.10-6 to 10-7. Civil Engineering and Development Department & Planning Department.
[51] Civil Engineering and Development Department & Planning Department. (2013). *Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report,* p.10-6 to 10-7. Civil Engineering and Development Department & Planning Department.
[52] Civil Engineering and Development Department & Planning Department. (2013). *Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report,* p.10-6 to 10-7. Civil Engineering and Development Department & Planning Department.

Table 11.1 Summary of Archaeological Background

Survey Area	Number of Test Pits	Number of Auger Holes	Sign of Archaeological Potential
Border Road	3	12	No
Ma Tso Lung	6	20	No
Ma Tso Lung Road	9	20	No

11.5 Built Heritage Impact Assessment

Introduction

11.5.1 The Appendix J of the EIA Study Brief (No. ESB-375/2025) stated that the Applicant shall conduct the BHIA on listed items as stated below.

i. declared monuments: the concerned place, building, site or structure has been declared a monument under the Antiquities and Monuments Ordinance (Cap. 53); and included in the "*List of Declared Monuments and Proposed Monuments (as at 10 October 2024)"*.

ii. proposed monuments: the concerned place, building, site or structure which is declared to be a proposed monument, proposed historical building, or proposed archaeological or palaeontological site or structure under the Antiquities and Monuments Ordinance (Cap. 53).

iii. List of Graded Historic Buildings, "List of Graded Historic Buildings (as at 26 June 2025)"

   a. grade 1 historic buildings: as defined by the AAB, they are buildings of outstanding merit, which every effort should be made to preserve if possible.

   b. grade 2 historic buildings: as defined by the AAB, they are buildings of special merit; efforts should be made to selectively preserve.

   c. grade 3 historic buildings: as defined by the AAB, they are buildings of some merit; preservation in some form would be desirable and alternative means should be considered if preservation is not practicable.

iv. List of New Items for Grading Assessment, "List of New Items for Grading Assessment" (as at 12 June 2025): currently under the latest *list of new items for grading assessment with grading results*, and yet to be accorded a grading by the AAB.

v. Government historic site identified by the Antiquities and Monuments Office: under the list of *"Government Historic Site Identified by AMO*" *(as at May 2022)*.

11.5.2 The listed built heritage are the tangibles (buildings/structures/features/places/areas) that have been identified by and / or accorded with a status by the Antiquities Authority. They are either under statutory protection or administrative protection by the Antiquities Authority.

11.5.3 There is no built heritage within the Project Boundary. However, within the 300m assessment area of the Project Boundary, there is one grade 2 historic building, namely *MacIntosh Fort (Ma Tso Lung)* (HB303), located approximately 35m away from the Project Boundary.

- *HB303 - MacIntosh Fort (Ma Tso Lung) (Grade 2 Historic Building)*

(**Appendix 11.1**: LMC-01 MacIntosh Fort (Ma Tso Lung) & **60722948/A10/FIGURE 11.4** refer)

11.5.4 *MacIntosh Fort (Ma Tso Lung)* is an observation post serving the purpose of illegal immigrant interception and border monitoring. It was built between 1949 and 1953[53] and ceased to operate in the 1980s[54].

11.5.5 The *MacIntosh Fort* is constructed of reinforce concrete and comprises a main structure and two single-storey structures. It is surrounded by chain link fencing and barbed wire. The structures are painted in green. One of the single-storey structure is connected to the main structure from the south, and its purpose is unknown. The remaining single-storey structure is located south of the main structures, and it served as a bathroom.

11.5.6 The plan of the main structure is generally in the shape of an isosceles obtuse triangle with rounded corners. Its apex is north facing while the two vertexes are south-east and south-west facing. The main structure is also surrounded by projected eaves. A two-storey circular (or octagonal) observation tower situated at the apex of the triangular plan. The rest of the main structure is of one storey. Battlemented parapets are installed above the roof and connected to the second storey of the observation tower, so that an observation terrace is formed. Its design could be classified as Modern Utilitarian[55].

Identification and Evaluation of Potential Impacts

- Construction Phase

11.5.7 Since *MacIntosh Fort (Ma Tso Lung) (HB303)* (grade 2 historic building) is located ~35m away from the Project Boundary, potential indirect impacts of ground borne vibration, tilting and settlement would be anticipated during the construction phase.

- Operation Phase

11.5.8 No direct or indirect impact is anticipated by the Project during the operation phase to *MacIntosh Fort (Ma Tso Lung)* (HB303).

Mitigation Measures

- Construction Phase

11.5.9 Potential indirect impacts, such as ground borne vibration, tilting and settlement, are anticipated during the construction phase, would affecting *MacIntosh Fort (Ma Tso*

[53] Antiquities Advisory Board. *Historic Building Appraisal MacIntosh Fort (Ma Tso Lung) Ma Tso Lung, Ta Kwu Ling, New Territories (Number 303)*. Antiquities Advisory Board.

[54] 陳天權（2019）。〈城市地標：殖民地時代的西式建築〉。香港：中華書局

[55] Antiquities Advisory Board (2010). Historic Building Appraisal – MacIntosh Fort (Ma Tso Lung). Antiquities Advisory Board https://www.aab.gov.hk/filemanager/aab/common/historicbuilding/en/303_Appraisal_En.pdf

Lung) (HB303). Monitoring of ground-borne vibration, tilting and ground settlement, is proposed to be employed.

11.5.10 *MacIntosh Fort (Ma Tso Lung) (HB303) (grade 2 historic building)* Monitoring of ground-borne vibration, tilting and ground settlement, is proposed to be employed for *MacIntosh Fort (Ma Tso Lung)* (HB303) during the site formation and construction phase. Any vibration and building movement induced from the construction works should be strictly monitored to ensure no disturbance and physical damages made to the heritage sites during the course of works.

11.5.11 The monitoring for HB303 should be incorporated with a set of Alert, Alarm and Action (3As) system following AMO's monitoring requirements for grade 2 historic building, as well as vibration-sensitive and dilapidated buildings against continuous vibration. The proposed 3As criteria for grade 2 historic building is presented in **Table 11.2**. The monitoring should be incorporated with a set of Alert, Alarm and Action (3As) system. The values for building settlement markers are clarified for monitoring both the building uplift and settlement in the vertical direction. The proposed 3As criteria are recommended in **Table 11.2** for reference.

11.5.12 Should the monitoring data be approaching to 3A levels, the project proponent shall propose and implement measures to mitigate movement situation at the heritage site. The project proponent should prepare a monitoring proposal with mitigation measures for each 3A levels exceeded for consideration by AMO, with examples, not limited to, increasing monitoring frequency, additional condition surveys, amendment / review of design of the construction, etc., so that the concerned built heritage would be protected and preserved.

11.5.13 A monitoring proposal, including type of monitoring, checkpoint locations, installation details, frequency of monitoring and response actions for each of the 3As level should be submitted to AMO for agreement before commencement of the works. Prior agreement and consent should be sought from the owner(s), stakeholder(s) and relevant Government department(s) for the installation of monitoring points on the built heritages before commencement of the works. AMO should be alerted in case any irregularities are observed.

Table 11.2 Proposed 3As Limiting Criteria for Vibration, Settlement, Tilting Monitoring for Cultural Heritage Resources

Alert, Alarm and Action (3As) Values			
Item	**Vibration (Peak Particle Velocity)**	**Settlement**	**Tilting**
MacIntosh Fort (Ma Tso Lung) HB303	5/6/7.5 mm/s	6/8/10 mm	1/2000, 1/1500, 1/1000

Note: Monitoring criteria would be subjected to review upon updates of grading status of heritage sites.

11.5.14 Meanwhile, due to the close proximity of HB303 and the Project Boundary, pre and post condition survey in construction phase is required. Condition survey and structural assessment should be carried out by an Authorized Person (AP) to inspect the physical condition and structural integrity of the building and provide advice on the subsequent mitigation measures based on the assessment result. Relevant professionals such as the Registered Structural Engineer (RSE) may also be involved to inspect the physical condition and structural integrity of the building. In case of the building structure being too vulnerable, structural strengthening measures may be

required upon agreement with the AMO, but the measures must be reversible and with minimal impact to the building fabric.

11.5.15 Regular and interval inspections or condition surveys may be considered if any irregularities are observed from the monitoring record, with circulation to AMO for record.

- Operation Phase

11.5.16 Since no direct or indirect impact is anticipated by the Project during the operation phase, no mitigation measure is required.

Summary

11.5.17 There is no built heritage within the Project Boundary. However, within the 300m assessment area of the Project Boundary, there is one grade 2 historic building, namely *MacIntosh Fort (Ma Tso Lung)* (HB303), located approximately 35m away from the Project Boundary. Potential indirect impacts, such as ground-borne vibration, tilting, and settlement, are anticipated during the construction phase, would affecting *MacIntosh Fort (Ma Tso Lung)* (HB303). Monitoring of ground-borne vibration, tilting and ground settlement, is proposed to be employed during construction phase. Meanwhile, pre and post condition survey in construction phase is required. Condition survey and structural assessment should be carried out by an Authorized Person (AP) to inspect the physical condition and structural integrity of the building and provide advice on the subsequent mitigation measures based on the assessment result.

11.5.18 Details of the proposed construction works within 50m distance of the above-mentioned, along with the impact assessment and mitigation measures when necessary, shall be submitted for AMO's consideration at the detailed design stage.

11.6 Archaeological Impact Assessment

Introduction

11.6.1 Archaeological heritage are the places/areas of archaeological significance, or where the archaeological potential is considered noteworthy. The archaeological heritage exclusively including the following:

i. **site of archaeological interest in Hong Kong**: under the *list of sites of archaeological interest in Hong Kong* maintained by the AMO (as at Nov 2012).

ii. area of archaeological potential identified in previous studies and / or this Project but not included in the item above.

Previous Archaeological Surveys

11.6.2 There is no SAI within the Project Boundary or its 300m assessment area.

11.6.3 Part of the area within the Project Boundary is covered by previous archaeological survey areas as detailed in **Section 11.4**, with archaeological potential assessed.

- Drainage Improvement in the Northern New Territories - Package B Archaeological Survey Final Report[56]

11.6.4 Archaeological findings dated to *Han* and early *Tang* dynasties were found in that survey, which is outside the 300m assessment area of this Project (**60722948/A10/FIGURE 11.3a** refers). The findings were "found as surface finds along the margin of two adjacent fish ponds on the alluvial plain at Ma Tso Lung" [57]. "The broad-scale disturbance of this area and conversion from paddy fields to fish ponds in the past 40 years has obscured much of the original landscape. Such disturbance may however throw up the traces of settlements and activity during prehistory and the early historic period…"[58]

11.6.5 The Han and early Tang artefacts identified within the marine deposit to the north of the assessment area represent secondary deposits that were transported to their present location through modern processes. Furthermore, the archaeological survey conducted in 2012 did not identify any such materials and concluded that there is no sign of archaeological potential[59]. As these finds are not *in situ* and were introduced into the deposit by non-archaeological means, their presence does not indicate any archaeological potential within the current Project Boundary.

- Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report [60,61]

11.6.6 As stated in **Section 11.4.23** an archaeological survey was conducted in the moderate archaeological potential area and partly at the margin of fishponds (**60722948/A10/FIGURE 11.3b** refers). It was located along the then-proposed Eastern and Western Connection Road. In particular, the survey areas along Border Road, Ma Tso Lung, and Ma Tso Lung Road are located in the vicinity of the Project Boundary of this Project. The surveyed areas were located in hill slopes, margin of fish ponds and alluvium[62]. The details of the survey findings are as follows.

i. Survey along Border Road

11.6.7 The auger holes tests conducted in this area all reached the regolith deposit. Meanwhile, test pits showed that a modern layer was between the surface layer and the regolith deposit.

ii. Survey at Ma Tso Lung

[56] Civil Engineering and Development Department & Planning Department. (2013). Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report. Civil Engineering and Development Department & Planning Department.
[57] Archaeo-Environments Ltd. (2004) Drainage Improvement in the Northern New Territories – Package B Agreement CE 5/2002(DS) Archaeological Survey Final Report, p.56. Archaeo- Environments Ltd.
[58] *Ibid.*
[59] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)
[60] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)
[61] Civil Engineering and Development Department & Planning Department. (2013). Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report. Civil Engineering and Development Department & Planning Department.
[62] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)

11.6.8 These tests reached water-soaked sludge or sandy deposits; or regolith deposit "soon after digging through the surface layer" [63].

iii. Survey along Ma Tso Lung Road

11.6.9 The tests either reached modern fill layers or regolith deposit.

11.6.10 That survey covered a large area in Ma Tso Lung and in vicinity of the Loop. And the field data concluded that there is no sign of archaeological potential for these areas (**60722948/A10/FIGURE 11.3c** refers).

Evaluation of Archaeological Potential

11.6.11 Although relatively flat areas were deemed suitable for past human settlement in both prehistoric and historical periods as mentioned in the abovementioned reports[64], the survey results in 2012[65] has repudiated the presumption. Except the modern fill layers found in some test pits, there was no cultural layer found between surface soil and regolith layer. The absence of cultural layer, as well as other archaeological findings, reflected that there is no sign of archaeological potential.

11.6.12 The Project Boundary and its 300m assessment area are located in the relative flat areas near the hill foot. They are located in the same area sharing similar environmental characteristics with the 2012 survey areas[66], which identified no archaeological potential. Even though the 2012 archaeological survey did not encompass the entire 300m assessment area and Project Boundary, the environmental conditions of the unsurveyed portions are consistent with those of the areas previously examined. On this basis, the conclusion of the 2012 survey, that there is no indication of archaeological potential, can reasonably be extended to the remaining areas of the current Project Boundary (**60722948/A10/FIGURE 11.5** refers).

11.6.13 This AIA with the conclusion of past surveys of those that concerns within the Project Boundary and its 300m assessment area of this project after reviewing all available archaeological information from past surveys. Hence, it is the opinion of this AIA that previous archaeological surveys have provided sufficient archaeological information for assessing the archaeological potential of the Project Boundary and its 300m assessment for this project.

Identification and Evaluation of Potential Impacts

11.6.14 There is no SAI within the Project Boundary or 300m assessment area. No impact is anticipated to any SAI.

11.6.15 Since no archaeological potential has been identified within the Project Boundary, no archaeological impact is anticipated by the Project. There is no archaeological impact.

[63] Civil Engineering and Development Department & Planning Department. (2013). Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report. Civil Engineering and Development Department & Planning Department.
[64] Civil Engineering and Development Department & Planning Department. (2013). Agreement No. CE53/2008 (CE) Planning and Engineering Study on Development of Lok Ma Chau Loop – Investigation EIA Report. Civil Engineering and Development Department & Planning Department.
[65] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)
[66] Hong Kong Institute of Archaeology (2012). *Planning and Engineering Study on Development of Lok Ma Chau Loop, Archaeological Field Survey Final Report*. (Contract No. FL/2011/01)

Mitigation Measures

11.6.16 There is no impact to any SAI. No mitigation measure is anticipated to any SAI.

11.6.17 Since there is no archaeological impact, no mitigation measure is required. As a precautionary measure, pursuant to the Antiquities and Monuments Ordinance (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

Summary

11.6.18 There is no SAI within the Project Boundary or 300m assessment area. Meanwhile, no archaeological potential has been identified within the Project Boundary. Therefore, no archaeological impact is anticipated from the Project and no mitigation measure are required.

11.7 Conclusion

11.7.1 There is no built heritage within the Project Boundary. However, within the 300m assessment area of the Project Boundary, there is one grade 2 historic building, namely MacIntosh Fort (Ma Tso Lung) (HB303). Potential indirect impacts, such as ground borne vibration, tilting and settlement, are anticipated during the construction phase, would affecting MacIntosh Fort (Ma Tso Lung) (HB303). Monitoring of ground-borne vibration, tilting and ground settlement is proposed during construction phase. Meanwhile, pre and post condition survey in construction phase is required. Condition survey and structural assessment should be carried out by an Authorized Person (AP) to inspect the physical condition and structural integrity of the building and provide advice on the subsequent mitigation measures based on the assessment result. Regular and interval inspections or condition surveys may be considered if any irregularities are observed from the monitoring record, with circulation to AMO for record.

11.7.2 There is no SAI within the Project Boundary or 300m assessment area. Meanwhile, no archaeological potential has been identified within the Project Boundary. Therefore, no archaeological impact is anticipated from the Project and no mitigation measure are required. As a precautionary measure, pursuant to the Antiquities and Monuments Ordinance (Cap. 53), the project proponent is required to inform the Antiquities and Monuments Office (AMO) immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

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